

Song-Ze Yu

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Research Interests

My research spans two parallel directions: grounded **music understanding** for audio-language models, and **controllable** music and audio-effect generation. My long-term goal is to build audio-language models with **real music intelligence**, offering musicians new interfaces to discuss, collaborate, or even interact with non-musical audiences.

Education

University of California, Berkeley

Visiting Student in Computer Science

• GPA: 3.9 / 4.0

Aug. 2025 - May 2026

Advised by Prof. Trevor Darrell and Carmine-Emanuele Cella

National Tsing Hua University

B.S. in Computer Science

• Overall GPA: 3.91 / 4.30

Sep. 2023 - Jun. 2026

Advised by Prof. Liu Yi-Wen

National Cheng Kung University

B.S. in Computer Science and Information Engineering

• Overall GPA: 4.16 / 4.30; highest department rank: 5 / 113

Sep. 2022 - Jun. 2023

Publications

(* indicates equal contribution; † indicates corresponding author)

- [1] Milan Liessens Dujardin*, **Song-Ze Yu***, Craver Corbyn Thomas-Smith, David M. Chan, Karina Nguyen, “*PitchBench: Measuring Pitch Hearing in Audio-Language Models*,” NeurIPS 2026 E&D.
- [2] **Song-Ze Yu**†, Milan Liessens Dujardin, Yuxuan Cai, Wantong Zhang, Brian Cruz, Jeremy Wagner, Carmine-Emanuele Cella†, “*InstructFX2FX: A Multi-Turn Text-to-Effect System for Sequential Audio Effect Refinement*,” DAFx 2026 Demo.
- [3] Milan Liessens Dujardin, **Song-Ze Yu**, Kevin Miao, “*MuSP-Bench: Advanced Multimodal Benchmarking of Music Understanding across Score and Performance*,” ISMIR LBD 2026.
- [4] **Song-Ze Yu**, Joseph Suh, Serina Chang, David M. Chan, “*Anamnesis: An Open-Source Platform for Large-Scale Backstory-Conditioned Survey Simulation*,” EMNLP 2026 Demo.

Manuscripts Under Review

- [5] **Song-Ze Yu**, Fang-Duo Tsai, Yi-Hsuan Yang, “*MIDI-REPA: Improving Text-to-Music Generation with Symbolic Representation Alignment*,” under review.
- [6] **Song-Ze Yu**, David M. Chan, “*UniFront: Revisiting Time-Frequency Frontends for Unified Music and Speech Modeling*,” under review.
- [7] Jun-You Wang, **Song-Ze Yu**, Xuanjun Chen, Chih-Kai Yang, Ke-Han Lu, Sung-Feng Huang, “*Diagnosing Music Understanding in Audio-Language Models: Encoder-Decoder Probing and Task Adaptation*,” under review.
- [8] Milan Liessens Dujardin, **Song-Ze Yu**, Kevin Miao, “*Unifying Score and Performance for Fine-Grained Music Understanding in Audio-Language Models*,” under review.

Research Experience

Music and AI Lab, National Taiwan University

Research Assistant

Jun. 2026 - Present

Advised by Prof. Yi-Hsuan Yang

- Developing **MIDI-REPA** to improve text-to-music generation by aligning audio-model representations with symbolic musical structure.
- Investigating music understanding in audio-language models through **encoder-decoder probing and task adaptation**.
- Leading a collaborative **music LALM** project with BAIR, NVIDIA, and Prof. Hung-yi Lee; model training is underway, connecting audio representations with language modeling.

Berkeley AI Research (BAIR) Lab, UC Berkeley

Researcher

Feb. 2026 - Present

Advised by David M. Chan and Prof. Trevor Darrell

- Develop **PitchBench**, a 28-experiment hierarchical benchmark showing that frontier audio-language models remain unreliable at both absolute and relative pitch perception, from isolated tones to melodic context.
- Built **Anamnesis**, an open-source survey-simulation platform that scales the **Anthology** method to 30K+ narrative backstories; reproduced Pew Research and New Yorker studies and better matched real-world opinion distributions than persona-prompting baselines.
- Developing **UniFront** to study time-frequency frontends for unified music and speech modeling; training audio encoders is underway.

Center for New Music and Audio Technologies Lab, UC Berkeley

Undergraduate Research Lead

Sep. 2025 - Jun. 2026

Advised by Prof. Carmine-Emanuele Cella

- Formulated **sequential FX refinement** as a stateful multi-turn problem and built **InstructFX2FX**, pairing LLM planning with CLAP-guided optimization to update effect chains without losing prior instructions.
- Unified score structure and aligned performance information to generate **fine-grained, time-grounded** music descriptions and question-answer pairs for audio-language models.

AHG Music Lab, National Tsing Hua University

Undergraduate Researcher

Jul. 2024 - May 2025

Advised by Prof. Yi-Wen Liu

- Developed the **VTR model**, a single-shot audio-to-EQ model trained on a ReaScript-generated piano dataset for tone replication (MSE: 0.0216); integrated it into a VST plugin at Positive Grid, then moved beyond single-shot control toward iterative effect refinement with **InstructFX2FX** at CNMAT.

Selected Engineering Experience

BerkeleyTime

Machine Learning Pod Lead

Sep. 2025 - May 2026

Spring 2026

- Interviewed and formed a **10-member ML Pod**, matching researchers and engineers by background into pairs across three workstreams: semantic-search maintenance, recommendation systems, and enrollment-probability prediction.
- Set weekly milestones, tracked progress, and reviewed code while coordinating researcher-engineer pairs with designers to bridge model selection, interface design, and production deployment.
- Led development of a **new Explore page** integrating the course-recommendation system into BerkeleyTime's user experience ([presentation](#)).

Machine Learning & Full-Stack Engineer, Catalog Pod

Fall 2025

- Built BGE-based semantic search as a FastAPI microservice backed by FAISS and deployed it behind a toggle with fuzzy search on the Catalog page for **122K+ monthly users**.
- Reduced GraphQL first-load latency from **25 seconds to 1 second** through pagination and Redis caching.
- Shipped **20+ pull requests** and resolved urgent on-call bugs involving cross-listed courses and ghost classes.

Positive Grid

Machine Learning Intern

Jun. 2025 - Aug. 2025

Taipei, Taiwan

- Served as Scrum Master for **Amp-AI-SaaS** and designed the **Multi-Agent Orchestration Harness** for software engineering across Positive Grid's audio products.
- Built **two JUCE-based VST plugins**, including one that integrated the **VTR model** through a PyTorch inference pipeline.
- Explored source separation and zero-shot virtual-amplifier modeling for music-industry applications.

Awards & Leadership

Claude × a16z × Berkeley M.E.T. Makeathon — Best Design

Oct. 2025

- Built Haven in five hours, a mobile app that personalizes home-design recommendations from room scans.

Meichu Hackathon — 2nd Place, Google Group

Oct. 2024

- Built [Travelity](#), an AI travel planner using interests, personality, and real-time map and search data.

Taipei Hackathon — 3rd Place

Sep. 2024

- Built [ParkFlow](#), a WebView microservice for real-time parking information, navigation, and notifications; selected for integration into Taipei's official city app (**3M+ downloads**) and awarded NT\$100,000.
- Produced a [hackathon vlog](#) with **1K+ views** and was [invited by the Taipei City Government](#) the following year to share the winning experience with Taipei residents.

SITCON Hackathon — Group Award Winner

Jul. 2024

- Built [ccDiary](#), an AI-assisted mental-health journal with emotion analysis and anonymous social features.

National Cheng Kung University — Campus Speaker

Aug. 2023

- Delivered an [invited talk](#) for incoming students on writing music as a diary, then turned audience-submitted phrases and sentences on Slido into improvised songs performed live.

Selected Music Experience

Pianist, Composer & Producer

[Music CV](#) | [YouTube \(7K subscribers\)](#)

- Performed **24 solo piano recitals** across Greater China and appeared on **10+ Chinese-language TV variety shows and New Year galas** (春晚) between 2009 and 2016.
- Shared the stage with **Jay Chou (周杰倫)** **three times** and appeared as a special guest at a concert by **Yan Jue (嚴爵)** at Taipei Arena, performing eight-hands piano with Bai-An (白安) and Cosmos People (宇宙人).
- Composed **20+ original songs with 20K+ cumulative views**; performs on piano/keys, lead vocal, cajon, and electric guitar, and arranges, mixes, and masters original work.